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Genomic Ledger Synthesis — SLO Execution Strategy

Sage Clokey — June 2026

What San Luis Obispo Has

Cal Poly SLO offers:

- **MS in Biomedical Engineering** (thesis track) — no PhD program
- **MS in Biological Sciences** — ~10 students/year, 2-year program, no PhD
- **Center for Applications in Biotechnology (CAB)** — 102,000 sq ft in the Frost Center, partnerships with industry, interdisciplinary biotech research
- **Cal Poly CIE HotHouse** — startup incubator, 125+ companies launched, med-tech focused vertical with lab space
- **REACH Central Coast** — regional innovation hub connecting med-tech, biotech, and life sciences

Applied Biotechnology Institute — a biotech company in Cal Poly Technology Park with a protein production platform.

The Challenge

SLO does not have a PhD program in synthetic biology, genome engineering, or anything close. Cal Poly is a teaching-focused polytechnic — strong undergrad/MS programs but not research-university infrastructure for GLS. No genome-scale metabolic modeling labs. No synthetic yeast. No cell-free genome assembly capability.

Option A: Remote PhD + SLO Base

Some programs allow remote/hybrid arrangements, especially for computational-heavy work:

- Do the computational side from SLO (compatibility engine, FBA modeling, block library design, bioinformatics pipeline)
- Travel to the advisor's lab for wet lab milestones (2-4 weeks at a time, 3-4 times/year)
- NYU, Harvard, and Stanford all have students who work partially remote post-COVID

- GLS work is ~60% computational, ~40% wet lab — this split makes hybrid feasible

Option B: Cal Poly MS + SBIR

1. Enroll in Cal Poly's **MS Biomedical Engineering** (thesis track)
2. Use CAB facilities for the computational and basic molecular biology work
3. Write thesis on the GLS compatibility engine + in silico validation
4. Simultaneously file SBIR Phase I through a spin-out entity
5. Use SBIR funding (75K) to access wet lab work through a **contract research organization (CRO)** or collaborate with a UC lab (UCSB is 2 hours south, has strong synbio)
6. Graduate in 2 years with an MS, a patent, and SBIR funding

Option C: Build the Company From SLO

1. Use **Cal Poly CIE HotHouse** for incubation — they support life science startups
2. Build the software stack from SLO (compatibility engine, shadow price calculator, block library management, consensus tracking)
3. Contract out wet lab validation to:
 - **Twist Bioscience** (SF) — DNA synthesis
 - **Ginkgo Bioworks** — foundry services for strain engineering
 - **ECU Bio / Strateos** — robotic cloud lab for automated screening
4. Apply for **NSF SBIR** or **DOE SBIR** — both fund from anywhere, no PhD required
5. Use REACH Central Coast network for local mentorship and connections

Option D: UCSB Collaboration (2 hours south)

UC Santa Barbara has:

- **Dept of Chemical Engineering** with metabolic engineering labs
- **Dept of Molecular, Cellular, and Developmental Biology**
- **California NanoSystems Institute (CNSI)** — shared instrumentation
- PhD programs that could host GLS wet lab work

Live in SLO, enroll at UCSB for a PhD, and commute 2x/week. Or establish a formal collaboration where Cal Poly MS work feeds into UCSB lab validation.

Recommended Path: Option B + C Combined

Timeline	Action
Fall 2026	Enroll Cal Poly MS BME, file provisional patent, start building software
Winter 2027	Apply SBIR Phase I, thesis proposal on GLS compatibility engine
Spring 2027	CIE HotHouse application, begin computational validation
Summer 2027	Contract Twist/Ginkgo for first wet lab blocks
Fall 2027	SBIR funding arrives, scale wet lab through CRO
Spring 2028	Defend MS thesis, company operational

This gets an MS, a patent, a funded company, and experimental data in **2 years** instead of 5-6 for a PhD. The PhD can come later if needed — or the company makes it unnecessary.

The software stack (compatibility engine, FBA shadow prices, block ordering, consensus tracking) is 100% buildable from SLO with a laptop. The wet lab comes through partnerships and CROs. You do not need to be in Boston or SF to start this.

Key SLO Resources

Resource	What It Provides
Cal Poly CAB (Frost Center)	102K sq ft biotech lab space, industry partnerships
Cal Poly CIE HotHouse	Startup incubation, 125+ companies launched
REACH Central Coast	Regional innovation network, mentorship
Applied Biotechnology Institute	Protein production platform in Cal Poly Tech Park
UCSB (2 hrs south)	PhD programs, CNSI shared instruments, metabolic engineering labs

Key Contacts to Make

Who	Why
Cal Poly BME Graduate Coordinator	MS admission, thesis advisor matching
Dr. Lily Laiho (BME Dept Chair)	Research alignment, lab access
Cal Poly CAB Director	Biotech lab access, industry connections
CIE HotHouse Program Manager	Incubation application, lab space
REACH Central Coast	Regional biotech network, mentorship
UCSB Chemical Engineering faculty	Potential collaboration for wet lab validation

Action Items

1. **File provisional patent** — USPTO, 00 micro-entity. Cover the compatibility engine, shadow-price-guided block ordering, and multi-node consensus validation. Do this before talking to anyone.
2. **Apply to Cal Poly MS BME** — thesis track, Fall 2026 or Winter 2027 start
3. **Contact Cal Poly CAB** — explore lab access and research partnership options
4. **Contact CIE HotHouse** — application timeline, what they need to see
5. **Scope SBIR** — NSF 26-503 (Biotechnology) or DOE (Biological & Environmental Research). Letter of intent usually due ~3 months before full proposal.
6. **Build the software** — the compatibility engine, FBA shadow price calculator, and block library manager can all be built now, from SLO, with Python and the tools already in the Adaptive Genome Design System.